These classical concepts are deeply ingrained in our thinking. They are the foundation of our instinctual understanding of the physical world, and it's very hard to get past them. But get past them we must, if we are to understand the quantum world.

6.3 COMBINING QUANTUM SYSTEMS

Charlie's two coins formed a single classical system, composed of two classical subsystems. Quantum mechanics also allows us to combine systems, as we found out in the Mathematical Interlude on tensor products (Section 6.1).

Alice and Bob have kindly agreed to provide a variant of the coin/die system they loaned us for the Interlude on tensor products. Instead of a coin and a die, the new system is built up from two spins, meaning that we'll have a chance to put our knowledge of single spins to work.

As before, we will sometimes use the oddball notation $|a\}$ to remind us that Alice's state-vectors are not in the same state-space as Bob's, and that we're not allowed to add them together. On the other hand, recall that each member of an orthonormal basis for S_{AB} is labeled by a pair of vectors, one from S_A and one from S_B. We will make frequent use of the notation $|ab>$ to label a single basis vector of the combined system. These doubly indexed basis vectors *can* be added together, and we'll be doing that a lot.

As we explained in the Interlude, labeling a basis vector with a pair of indices takes some getting used to. You should think of the pair *ab* as a single index labeling a single state.

Let's look at an example. Consider some linear operator **M** acting on the space of states of the composite system. As usual, it can be represented as a matrix. The matrix elements are constructed by sandwiching the operator between basis vectors. Thus, the matrix elements of **M** are expressed as

$$\langle a'b'|\mathbf{M}|ab\rangle = M_{a'b',ab}.$$